

U2-AT2 PROBLEM

1. **PROBLEM STATEMENT:**

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic, where roads may suddenly become inaccessible due to flooding, and weather conditions affect the travel cost. The drone has access to a heuristic estimate (straight-line distance), partial real-time updates about blocked paths, and variable movement costs depending on terrain and weather. The objective is to reach the destination with minimum delay while ensuring safe and efficient navigation.

2. **PROBLEM STATEMENT:**

A smart city implements an AI-based traffic management system to optimize traffic signal timings across hundreds of intersections. The main objectives are to minimize total vehicle waiting time, fuel consumption, and traffic congestion. Since traffic conditions continuously change throughout the day, the AI system must dynamically adjust signal timings. The search space is extremely large, and traditional search methods are inefficient because the system may become trapped in locally optimal solutions. Therefore, optimization algorithms are required to find efficient traffic signal configurations.

3. **PROBLEM STATEMENT:**

An autonomous rover is deployed on Mars to explore unknown terrain. The rover must safely navigate to collect scientific samples while avoiding hazards such as rocks and craters. It operates with limited sensing capability and does not have a complete map of the environment. As it explores, the rover continuously updates its knowledge about the surroundings and makes real-time decisions. Since the environment is uncertain and only partially known, the rover requires an intelligent search strategy that can adapt while moving.

4. **PROBLEM STATEMENT:**

A university plans to develop an AI-based exam timetable generation system. The system must automatically schedule examinations while satisfying several constraints, including ensuring that no student has overlapping exams, limited availability of exam halls, subject precedence requirements, faculty availability, and special accommodations for certain students. As the number of courses, students, and constraints increases, the scheduling process becomes highly complex. Therefore, the problem is formulated as a **Constraint Satisfaction Problem (CSP)** and solved using Backtracking Search with Constraint Propagation.

5. **PROBLEM STATEMENT:**

A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players by making intelligent decisions within strict time limits. Each game state has multiple possible moves, creating a very large decision tree that is computationally expensive to evaluate. The objective is to choose the best move while maintaining a balance between optimal decision-making and real-time performance. To solve this problem, the AI uses the **Minimax algorithm** with **Alpha-Beta Pruning** to reduce unnecessary computations.